

- Compliant with NIST FIPS publication 197 “Advanced encryption standard (AES)” (November 2001)
- Encryption and decryption with multiple chaining modes:
 - Electronic codebook (ECB) mode
 - Cipher block chaining (CBC) mode
 - Counter (CTR) mode
 - Galois counter mode (GCM)
 - Galois message authentication code (GMAC) mode
 - Counter with CBC-MAC (CCM) mode
- 128-bit data block processing, supporting cipher key lengths of 128-bit and 256-bit
 - 51 or 75 clock cycle latency in ECB mode for processing one 128-bit block with, respectively, 128-bit or 256-bit key
- Integrated key scheduler to compute the last round key for ECB/CBC decryption
- 256-bit write-only registers for storing cryptographic keys (eight 32-bit registers)
- 128-bit registers for storing initialization vectors (four 32-bit registers)
- 32-bit buffer for data input and output
- Automatic data flow control supporting two direct memory access (DMA) channels, one for incoming data, one for processed data. Only single transfers are supported.
- Data-swapping logic to support 1-, 8-, 16-, or 32-bit data
- AMBA AHB slave peripheral, accessible through 32-bit word single accesses only. Other access types generate an AHB error, and other than 32-bit writes may corrupt the register content.
- Software (in CPU mode only, not in DMA mode) can suspend a message if AES needs to process another message with a higher priority, then resume the original message.

Table 5. AES features

Modes or features ⁽¹⁾	AES	SAES
ECB, CBC chaining	X	X
CTR, CCM, GCM chaining	X	X
AES 128-bit ECB encryption in cycles	51	480
DHUK and BHK key selection	-	X
Resistant to side-channel attacks	-	X
Shared key between SAES and AES	X	
Key sizes in bits	128/256	128/256

1. X = Supported.

3.23 Secure AES coprocessor (SAES)

The secure AES coprocessor (SAES) encrypts or decrypts data in compliance with the advanced encryption standard (AES) defined by NIST. It incorporates protection against side-channel attacks (SCA), including differential power analysis (DPA), certified SESIP, and PSA security assurance level 3.

SAES supports ECB, CBC, CTR, GCM, GMAC, and CCM chaining modes for key sizes of 128 or 256 bits, as well as special modes such as hardware secret key encryption/decryption (wrapped-key mode) and key sharing with a faster AES peripheral (shared-key mode).

SAES can load STM32 hardware secret master keys (boot hardware key BHK and derived hardware unique key DHUK) by hardware, usable but not readable by the application.

The peripheral supports DMA single transfers for incoming and outgoing data (two DMA channels are required). It is hardware-linked with the true random number generator (RNG) and the AES peripheral.

The SAES main features are: